

BEYOND TEXTUAL CHAIN-OF-THOUGHT: JEPa-CONDITIONED LATENT REASONING FOR LARGE AUDIO LANGUAGE MODELS

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ABSTRACT

Explicit textual Chain-of-Thought (CoT) has improved the reasoning ability of large audio language models (LALMs). However, textual CoTs are often constructed from text captions of audio and provide limited access to the acoustic evidence, which can introduce problems like hallucination. To address this modality-gap issue, we introduce JELAR, a Joint-Embedding Predictive Architecture (JEPa)-based latent reasoning framework that conditions latent reasoning supervision on acoustic representations learned from raw waveforms. During training, a frozen WavJEPa model provides representations learned directly from raw waveforms. A non-causal expert first constructs answer-aware queries, which cross-attend to WavJEPa embeddings to produce latent reasoning targets. The LALM is trained to predict these targets before generating its response. Experimental results show that JELAR improves the Audio-Reasoner baseline by 2.70 and 9.10 absolute percentage points on MMAU-mini and MMAR, respectively, demonstrating the effectiveness of JEPa-conditioned latent reasoning as an alternative to explicit textual CoT supervision.

Index Terms— Large Audio Language Model, Latent Reasoning, Chain-of-Thought, WavJEPa, Audio Reasoning

1. INTRODUCTION

Large audio language models (LALMs) have recently demonstrated increasingly strong capabilities in speech understanding, spoken dialogue, environmental sound analysis, music perception, and audio question answering [1, 2, 3]. Nevertheless, reasoning over audio remains substantially more challenging than recognizing isolated acoustic events [4, 5]. Complex tasks often require a model to identify relevant evidence, integrate information across time, relate sound sources, and combine acoustic perception with semantic inference [6, 7, 8].

Explicit chain-of-thought (CoT) has emerged as a promising approach to improving such capabilities [7, 8]. Audio-

Reasoner [8], for example, constructs structured textual reasoning traces and trains an LALM to generate a CoT before its final response. However, recent studies have highlighted a fundamental modality-gap concern in audio reasoning: as such CoTs are often synthesized from textual captions or other text-based descriptions rather than directly from the acoustic signal, their alignment with the underlying audio evidence is not explicitly ensured [5, 6]. This mismatch may contribute to hallucinations. This concern is particularly relevant to audio, where task-critical information may reside in prosody, timbre, overlapping events, temporal structure, background acoustics, and cross-source relationships that cannot always be faithfully reduced to a textual description.

Latent reasoning provides an alternative to this text-centric paradigm by allowing reasoning to take place in a continuous representation space [9, 10]. Yet latent reasoning alone does not address the modality-gap issue: without an appropriate acoustic target space, the learned latent reasoning trajectory may still encode predominantly linguistic or shortcut information. Therefore, the key challenge is not simply to move reasoning from text into a continuous space, but to construct latent supervision that incorporates representations derived directly from the acoustic signal, rather than relying solely on textual reasoning traces. However, many audio encoders used in LALMs, such as Whisper [11], are optimized with audio-to-text prediction objectives [12, 13]. Although effective for language-conditioned audio understanding, their representations are shaped toward linguistic prediction and may not provide an ideal target space for supervising acoustically conditioned latent reasoning.

This motivates using an acoustic representation space learned without text-prediction supervision. Joint-Embedding Predictive Architectures (JEPAs) learn representations by predicting latent targets from contextual observations, rather than reconstructing raw inputs or predicting textual labels. WavJEPa [14] extends this predictive objective to raw waveforms and captures temporally predictable acoustic structure without transcript, caption, or answer supervision. Such predictive waveform representations provide a promising space

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for constructing latent reasoning trajectories.

Based on this insight, we introduce **JELAR**, a **JEPA**-conditioned **Latent Audio Reasoning** framework for LALMs. During training, a frozen WavJEPA encoder extracts audio representations from the input waveform. A non-causal expert uses the question and ground-truth answer to construct a set of queries, which cross-attend to the WavJEPA embeddings to produce fixed-length latent reasoning targets. At inference, the LALM independently generates the fixed-length latent reasoning sequence and subsequently produces the textual answer. Experiments on MMAU-mini and MMAR demonstrate that JELAR improves average accuracy over the Audio-Reasoner baseline by 2.70 and 9.10 absolute percentage points. Our contributions are summarized as follows:

- We introduce JELAR, a latent reasoning framework that replaces explicit textual CoT supervision with continuous latent targets conditioned on JEPA-based waveform representations.
- We incorporate WavJEPA acoustic representations into a training-only non-causal expert to construct acoustically informed latent targets, which the LALM learns to autoregressively predict without the expert or WavJEPA at inference.
- Controlled experiments on MMAU-mini and MMAR show consistent gains over explicit CoT training, while ablations isolate the contributions of acoustic conditioning, WavJEPA-based targets, and latent-only reasoning.

2. METHOD

2.1. Overview

The architecture of the proposed JELAR is shown in Figure 1. Given an audio waveform a and a question q , the native audio encoder and the projector of the LALM first convert the waveform into a sequence of audio tokens:

$$X_a = E_A(a), \quad X_a \in \mathbb{R}^{N_x \times d_{LLM}}, \quad (1)$$

where N_x is the number of audio tokens and d_{LLM} is the input embedding dimension of the language model.

The LALM receives the audio tokens X_a together with the question tokens and generates a fixed-length sequence of continuous latent embeddings:

$$\hat{Z} = \{\hat{z}_1, \dots, \hat{z}_K\}, \quad \hat{z}_t \in \mathbb{R}^{d_{LLM}}, \quad (2)$$

where K denotes the latent reasoning steps. Each predicted latent embedding is fed back to the language model as the input embedding of the next reasoning step. During training, the latent inputs for subsequent reasoning steps are provided by a non-causal expert rather than by the model’s previous predictions. At inference, the model recursively feeds back its own predicted latent embeddings.

2.2. Supervised Fine-Tuning (SFT) for Latent Reasoning

Since continuous reasoning trajectories are not available in the training data, we introduce a non-causal expert, consisting of a bidirectional BERT encoder and a cross-attention module, to construct latent reasoning targets. For each training example (a, q, y) , the expert first forms answer-aware semantic queries from (q, y) , which then cross-attend to waveform representations extracted by WavJEPA [14].

JEPA-conditioned latent target construction: To obtain the acoustic representations, we use a frozen WavJEPA encoder E_J to convert the waveform into a sequence of embeddings [14]:

$$Z_a = E_J(a), \quad Z_a \in \mathbb{R}^{N_a \times d_J}, \quad (3)$$

where N_a and d_J denote the sequence length and feature dimension of the WavJEPA representations, respectively. To determine which acoustic information is relevant to the target answer, we first construct K answer-aware semantic queries by a BERT encoder [15]. Let $S_0 \in \mathbb{R}^{K \times d_B}$ denote K learnable token slots, where d_B is the hidden dimension of the bidirectional BERT. The question and answer embeddings are processed jointly with these slots:

$$R = \text{BERT} \left([S_0; W_q(\text{Emb}(q)); W_y(\text{Emb}(y))] \right)_{1:K}, \quad (4)$$

where W_q and W_y project the question and ground-truth answer embeddings into the BERT dimension and the semantic queries $R = \{r_1, \dots, r_K\}$ consist of the output states corresponding to the learnable slots. The semantic queries subsequently attend to the WavJEPA representation sequence through a cross-attention module:

$$G = \text{CrossAttn}(Q = R, K = W_J Z_a, V = W_J Z_a), \quad (5)$$

where W_J projects the WavJEPA representations into the query dimension. The representations are finally projected into the LALM embedding space:

$$Z^* = \{z_1^*, \dots, z_K^*\} = W_{\text{out}} G, \quad Z^* \in \mathbb{R}^{K \times d_{LLM}}. \quad (6)$$

These embeddings constitute the JEPA-conditioned latent targets, combining answer-aware semantic queries with acoustic information retrieved from the predictive waveform representations.

Training objective. During teacher-forced training, Z^* is inserted between $\langle \text{LAT} \rangle$ and $\langle / \text{LAT} \rangle$ tokens. Let f_{lat} be a latent prediction head. The hidden state at the $\langle \text{LAT} \rangle$ position predicts the first latent embedding, while the hidden state at the position of z_{t-1}^* predicts the next one:

$$\hat{z}_t = f_{\text{lat}} \left(h_{z_{t-1}^*} \right), \quad t = 2, \dots, K. \quad (7)$$

We optimize the latent predictions using Smooth L1 loss:

$$\mathcal{L}_{\text{lat}} = \frac{1}{K} \sum_{t=1}^K \text{SmoothL1}(\hat{z}_t, \text{sg}(z_t^*)). \quad (8)$$

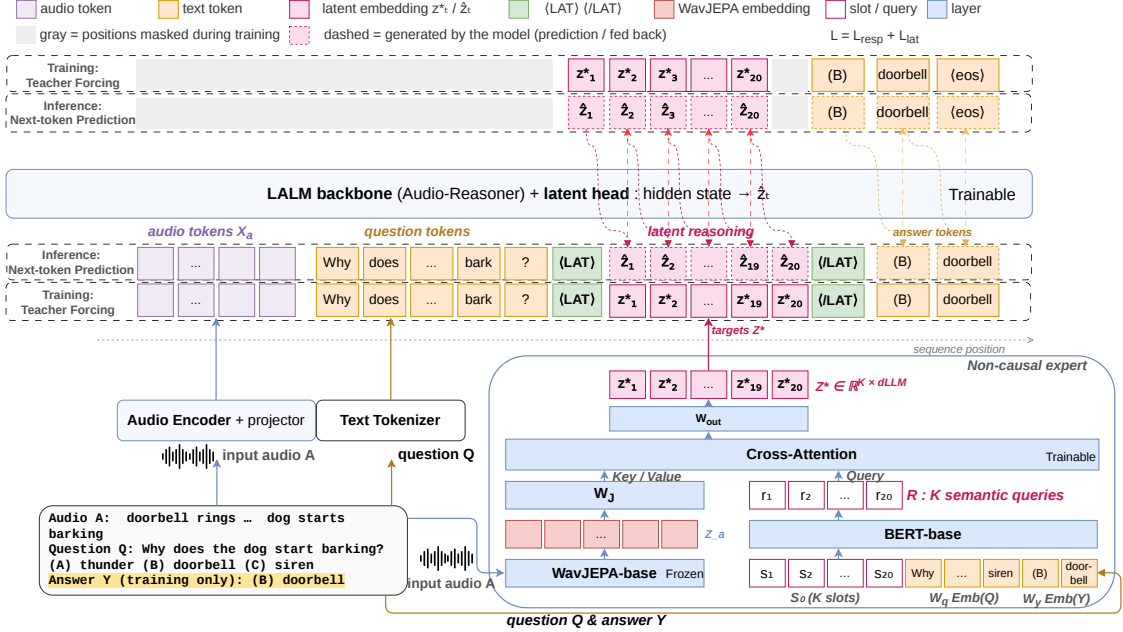


Fig. 1. The training and inference pipeline of proposed JELAR.

Stop-gradient is applied to the latent target in Eq. (8).

The response loss is computed over the ground-truth answer tokens:

$$\mathcal{L}_{\text{resp}} = -\frac{1}{|y|} \sum_{i=1}^{|y|} \log p_{\theta}(y_i | X_a, q, Z^*, y_{<i}). \quad (9)$$

Z^* is inserted into the LALM without detachment, thus Eq. (9) also optimizes the non-causal expert, encouraging it to construct latent targets that support response generation.

The complete training objective is

$$\mathcal{L} = \mathcal{L}_{\text{resp}} + \mathcal{L}_{\text{lat}}. \quad (10)$$

The WavJEPa encoder remains frozen, while all other components, including the non-causal expert, latent prediction head, audio encoder and projector, and LLM backbone, are jointly optimized.

2.3. Inference

At inference time, the non-causal expert and WavJEPa encoder are discarded. Given an audio waveform a and a question q , as well as a special token $\langle \text{LAT} \rangle$, the LALM first generates the latent reasoning embeddings recursively. At step t , the final hidden state is mapped to a continuous latent embedding $\hat{z}_t$, which is fed back to LLM at the next time step $t + 1$. After K latent steps, the $\langle \text{LAT} \rangle$ token is appended. The LALM then continues standard autoregressive decoding to generate the formatted textual response:

$$\hat{y}_i \sim p_{\theta}(\cdot | X_a, q, \hat{Z}, \hat{y}_{<i}). \quad (11)$$

3. EXPERIMENTS

3.1. Experimental Setup

Model configuration. We build JELAR on the same Audio-Reasoner backbone as the baseline [8]. The explicit-CoT baseline and JELAR use the same training data and optimization configuration. The key difference is the reasoning supervision: Audio-Reasoner is trained with explicit textual CoT, whereas JELAR replaces textual CoT with continuous latent reasoning. We therefore emphasize their controlled comparison to assess the effect of replacing explicit CoT with JEPa-conditioned latent reasoning. In JELAR, a frozen WavJEPa-base encoder¹ provides the representations used to construct the latent targets, while the answer-aware query constructor is initialized from BERT-base-uncased². Unless otherwise specified, the latent reasoning stage contains $K = 20$ embeddings. We choose this based on performance on a held-out validation split of the development data.

We evaluate the models on MMAU-mini [17] and MMAR [16]. MMAU-mini consists of closed-choice questions covering sound, music, and speech. MMAR contains human-curated multiple-choice questions designed for **multi-step audio reasoning**.

3.2. Main Results

MMAU-mini results. As shown in Table 2, JELAR improves the average accuracy of Audio-Reasoner from 59.70%

¹<https://huggingface.co/labhamlet/wavjepa-base>

²<https://huggingface.co/google-bert/bert-base-uncased>

Table 1. Accuracy (%) on MMAR. Results marked with † are taken from the MMAR benchmark paper [16]; S, M, and Sp denote sound, music, and speech, respectively. Boldface denotes the better-performing result only within each controlled comparison.

Model	S	M	Sp	S-M	S-Sp	M-Sp	S-M-Sp	Avg.
Qwen2-Audio-Instruct†	33.33	24.27	32.31	9.09	31.19	30.49	25.00	30.00
Qwen-2.5-Omni-7B†	58.79	40.78	59.86	54.55	61.93	67.07	58.33	56.70
GPT-4o Audio†	53.94	50.97	70.41	63.64	72.48	62.20	75.00	63.50
Gemini 2.0 Flash†	61.21	50.97	72.11	81.82	72.48	65.85	70.83	65.60
Audio-Reasoner	44.24	35.92	46.26	27.27	47.71	50.00	25.00	43.70
JELAR	53.33	49.51	51.70	45.45	57.34	53.66	50.00	52.80

Table 2. Accuracy (%) on MMAU-mini. Results marked with † are reported in [8]. The bottom two rows constitute the primary controlled comparison.

Model	Sound	Music	Speech	Avg.
GPT-4o + Caption†	63.36	60.77	53.15	57.30
Gemini-1.5-Pro†	56.75	49.40	58.55	54.90
Qwen2-Audio-Instruct†	54.95	50.98	42.04	49.20
Audio-Reasoner	66.86	61.57	51.37	59.70
JELAR	64.31	62.28	60.66	62.40

Table 3. Effect of the latent reasoning length K . We report average classification accuracy (%) on MMAU-mini and MMAR.

K	8	12	16	20	24	28	32
MMAU-mini	60.60	61.00	61.80	62.40	60.00	61.10	60.80
MMAR	49.60	50.40	51.00	52.80	51.00	51.30	51.70

to 62.40%. The gain is mainly driven by the speech domain, which increases from 51.37% to 60.66%, while music improves slightly and sound accuracy decreases.

MMAR results. Table 1 shows that JELAR delivers a substantially larger gain on MMAR, raising average accuracy from 43.70% to 52.80%, an absolute improvement of 9.10 percentage points. Notably, it outperforms Audio-Reasoner across all seven categories, with especially large gains on mixed-domain reasoning tasks. The gain on MMAR is considerably larger than that on MMAU-mini. This is consistent with the design of MMAR, which places greater emphasis on multi-step reasoning and mixed-domain audio reasoning. The larger gain on MMAR is also consistent with the hypothesis that the proposed latent reasoning may be particularly beneficial for tasks requiring integration of multiple acoustic cues.

3.3. Ablation study

Table 3 examines the effect of latent reasoning length. Increasing K from 8 to 20 improves MMAU-mini from 60.60% to 62.40% and MMAR from 49.60% to 52.80%, with both benchmarks peaking at $K = 20$. Increasing K further does not yield additional gains, suggesting that more latent reasoning steps are not inherently beneficial once sufficient reason-

Table 4. Ablation of acoustic conditioning and reasoning format.

Variant	MMAU	MMAR
JELAR (WavJEPA, latent-only)	62.40	52.80
w/o acoustic conditioning, latent-only	60.50	47.60
Whisper encoder, latent-only	61.60	50.70
WavJEPA, latent + explicit CoT	61.40	49.30

ing capacity is reached.

Table 4 analyses the contributions of acoustic conditioning and reasoning format. Removing acoustic conditioning reduces the accuracy to 60.50% on MMAU-mini and 47.60% on MMAR. Conditioning the latent targets on the native Whisper encoder of Audio-Reasoner [11, 8] improves the results to 61.60% and 50.70%, while WavJEPA further raises them to 62.40% and 52.80%. These results indicate that acoustic conditioning provides additional benefit beyond answer-aware latent supervision, while WavJEPA provides a more effective acoustic target representation than the native Whisper encoder in this setting.

For reasoning format, explicit textual CoT provides no additional benefit once latent reasoning is introduced. Adding an explicit CoT stage after the latent sequence lowers performance from 62.40% to 61.40% on MMAU-mini and from 52.80% to 49.30% on MMAR. This supports latent-only reasoning as the more effective reasoning format in our setting.

4. CONCLUSION

In this work, we introduced JELAR, a JEPA-conditioned latent reasoning framework for LALMs. JELAR replaces explicit textual CoT supervision with continuous latent targets constructed by answer-aware semantic queries attending to frozen WavJEPA representations. Under controlled comparisons with Audio-Reasoner, JELAR improves average accuracy by 2.70 and 9.10 percentage points on MMAU-mini and MMAR, respectively. Ablation studies further support the contributions of acoustic conditioning, WavJEPA-based target representations, and latent-only reasoning. Future work will investigate larger backbones and more diverse audio reasoning data.

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